Basic setup and installation

1 Open a shell

1.1 Windows

1.2 macOS

1.3 Linux

2 Install Julia

2.1 Windows

Install Julia from the Windows store, found here:
https://www.microsoft.com/store/apps/9NJNWW8PVKMN
or using the following command in the shell:

```
winget install julia -s msstore in a terminal
```

2.2 Linux and macOS

Install Julia by entering the following command in your shell.

```
curl -fsSL https://install.julialang.org | sh
```

3 Install the required packages

3.1 Activate the Julia environment

If installing into the global environment, start the global julia environment in your shell:

```
julia
```

If installing into a project environment, navigate to the project directory first in your shell and then start a Julia REPL session using the `--project` flag:

```
cd ~/projects/paleodb
julia --project=.
```

Global installs are convenient, with packages available to any Julia program you run. Some workflows benefit from a "heavy" global environment with most common packages installed there `DataFrames.jl`, `CSV.jl`, `XLSX.jl`, `Makie.jl`, `Distributions.jl` and universally available to every project, with project-specific installs limited to specialized needs.

On the other hand, the lack of isolation between global and project-specific environments can result in globally-installed packages leaking into project tests, dependency chains, and so on. Workflows incorporating software engineering best practices typically benefit from having light global environments and heavy project environments: little to no packages installed globally, with all packages installed into project specific environments.

3.2 Import the package manager and confirm environment status

Either way, confirm that you are running in the correct environment by using the `status` function of the Julia package manager, `Pkg`:

```
using Pkg
Pkg.status()
```

A typical output if the global environment is active might be:

```
Status `~/.julia/environments/v1.12/Project.toml`
  [5fb14364] OhMyREPL v0.5.32
  [295af30f] Revise v3.11.0
Info Packages marked with  have new versions available and may be upgradable.
```

The first line of output from the `Pkg.status()` function is the path to the active Julia project file, and here, the path is to the global Julia environment.

```
Status `~/.julia/environments/v1.12/Project.toml`
```

If a project environment is active, on the other hand, will hand you would see the path to the project directory.

```
Status `~/projects/learning/juliacamp/paleobiology`
```

3.3 Install the basic packages

```
Pkg.add([
    "PaleobiologyDB",
    "DataFrames",
    "Statistics",
    "CairoMakie",
])
```